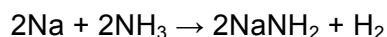


Section A

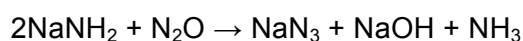
For each question there are four possible answers, **A**, **B**, **C** or **D**. Choose the **one** you consider to be correct.

- 1 Sodium azide, NaN_3 is the gas-forming component in many car airbag systems. The common synthesis method for NaN_3 is called the "Wislicenus process", which involves two steps starting from ammonia.

In the first step, ammonia is converted to sodium amide:



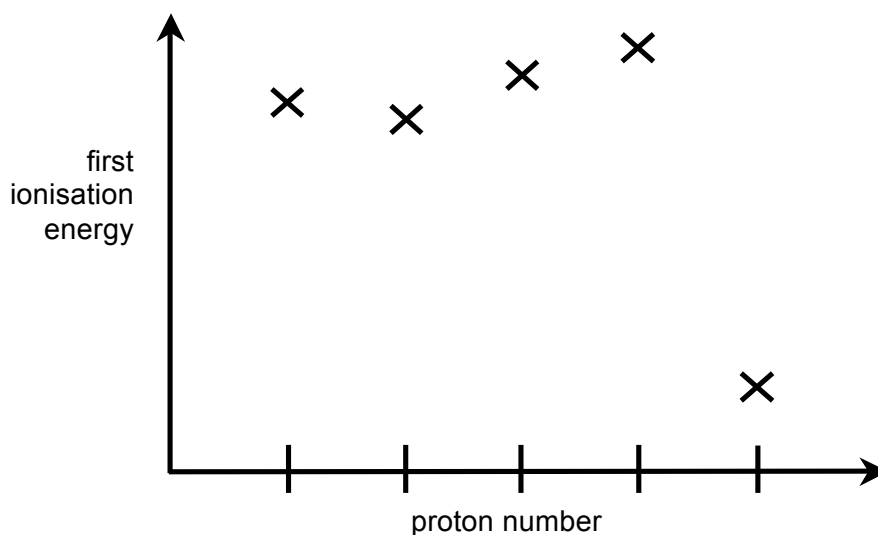
The sodium amide is subsequently combined with nitrous oxide:



What is the percentage yield of NaN_3 when 8.54 g of NaN_3 was produced from 10.1 g of Na?

- A** 60.0% **B** 65.0% **C** 70.0% **D** 75.0%

- 2 The first ionisation energy of five successive elements in the periodic table has the trend shown in the diagram.



What could be the second element of this sequence?

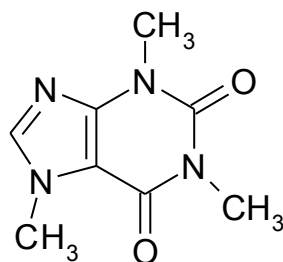
- A** N **B** O **C** F **D** Ne

- 3 Which formula represents a particle with the composition 1 proton, 2 neutrons and 0 electron?

(T represents tritium, ${}^3_1\text{H}$)

- A H^+ B He C T D T^+

- 4 Caffeine as shown below is found in varying quantities in the seeds, leaves and fruit of some plants, where it acts as a natural pesticide that paralyzes and kills certain insects feeding on the plants, as well as enhances the reward memory of pollinators.



How many lone pairs of electrons are present in this molecule?

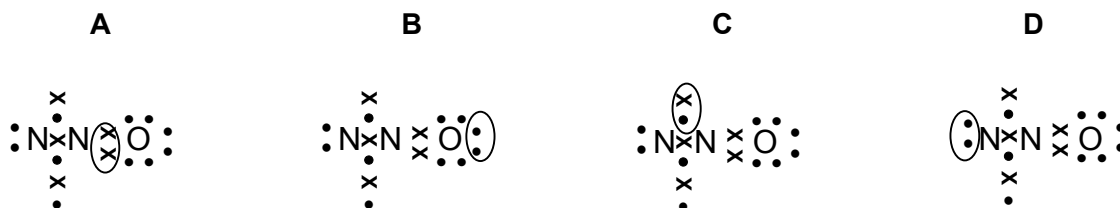
- A 2 B 4 C 6 D 8

- 5 Which of the following involves an increase in oxidation number of an element?

- A $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$
 B $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
 C $4\text{Ti}^{3+} + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{TiO}^{2+} + 4\text{H}^+$
 D $2\text{H}_3\text{PO}_4 + 3\text{Ba}(\text{OH})_2 \rightarrow \text{Ba}_2(\text{PO}_4)_3 + 3\text{H}_2\text{O}$

- 6 Dot and cross diagrams of dinitrogen oxide are shown below.

Which pair of electrons represents a coordinate bond?



7 Consider the following four compounds

- 1 $(\text{CH}_3)_3\text{CH}$
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- 4 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

What is the order of increasing boiling point of the compounds?

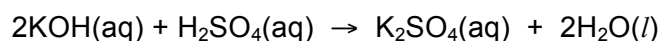
- A 1 → 2 → 3 → 4
- B 2 → 3 → 1 → 4
- C 1 → 4 → 3 → 2
- D 4 → 2 → 3 → 1

8 Silicon carbide (carborundum) is a shiny, hard, chemically inert material with a very high melting point. It can be used to sharpen knives and make crucibles.

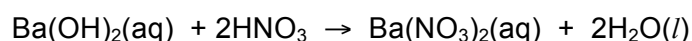
Which type of structure explains these properties?

- A a giant structure with covalent bonds between silicon and carbon atoms
- B a giant structure containing metallic bonding
- C a giant layer structure with covalent bonds between atoms and van der Waals' forces between the layers
- D a simple molecular structure with covalent bonds between the atoms of silicon and carbon

9 The heat liberated in the neutralisation given below is -114 kJmol^{-1} .



By using the information, what is the most likely value for the heat liberated in the following neutralisation?



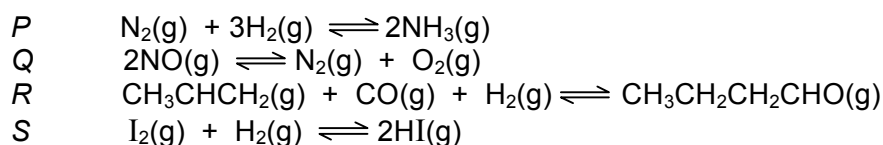
- | | |
|----------------------------|-----------------------------|
| A -29 kJmol^{-1} | B -57 kJmol^{-1} |
| C -86 kJmol^{-1} | D -114 kJmol^{-1} |

- 10** The lattice energies of rubidium fluoride, RbF, and caesium chloride, CsCl, are -760 kJmol^{-1} and -650 kJmol^{-1} respectively.

What is the lattice energy caesium fluoride, CsF, likely to be?

- A** -620 kJmol^{-1} **B** -720 kJmol^{-1}
C -800 kJmol^{-1} **D** -900 kJmol^{-1}

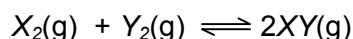
- 11** Four equilibria are shown.



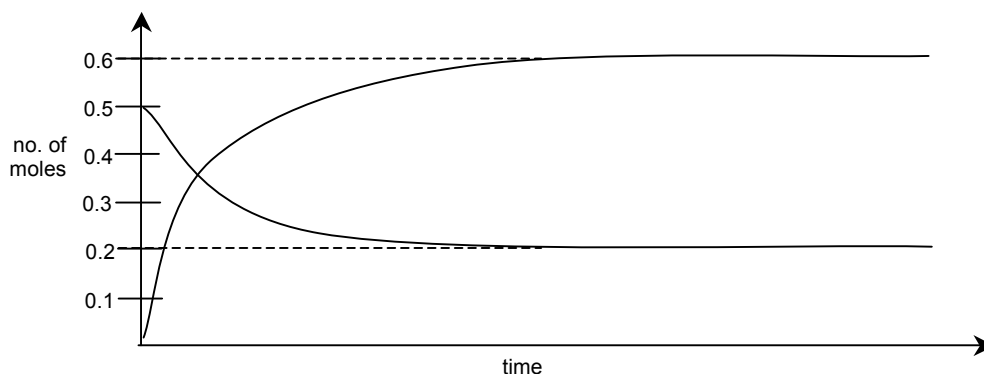
For which pair of equilibria does the position of equilibrium remain unchanged with an expansion to the volume of the reaction vessel?

- A** *P* and *Q* **B** *Q* and *R*
C *Q* and *S* **D** *R* and *S*

- 12** Two diatomic gases, X_2 and Y_2 react as follows:



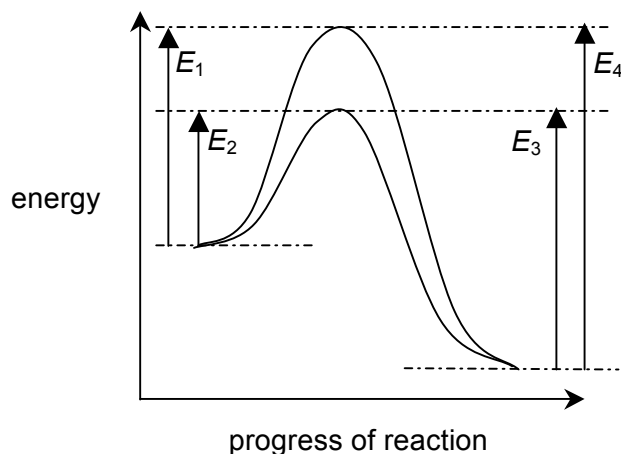
A mixture containing 0.5 moles of each X_2 and Y_2 is heated in a closed container and the reaction allowed to reach equilibrium. The graph shows how the number of moles of each gas varies with time.



What is the value of the equilibrium constant, K_c for this reaction?

- A** 3.0 **B** 9.0
C 12.0 **D** 15.0

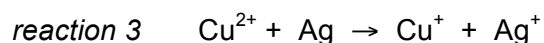
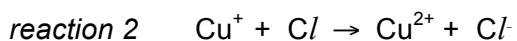
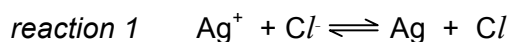
- 13 The energy diagram represents a reaction that occurs in the absence, and in the presence, of a catalyst.



Which of the following statements is correct?

- A** The enthalpy change of reaction is $(E_2 - E_3)$.
- B** E_4 is the activation energy for the reverse catalysed reaction.
- C** The forward reaction, with catalyst, is endothermic.
- D** The enthalpy change of reaction is reduced by using a catalyst.
- 14 Photochromic glass, used for sunglasses, darkens when exposed to bright light and becomes more transparent again when the light is less bright. The depth of the colour of the glass is related to the concentration of silver atoms.

The following reactions are involved.



Which statement about these reactions is correct?

- A** Cu^+ ion act as a catalyst.
- B** Cl^- ions act as an oxidising agent in reaction 1.
- C** Reaction 2 is the one in which light is absorbed.
- D** Ag atoms are reduced in reaction 3.

- 15** An element of the third period (Na to S) is heated in chlorine. The product is purified and then added to water. The resulting solution is found to be neutral.

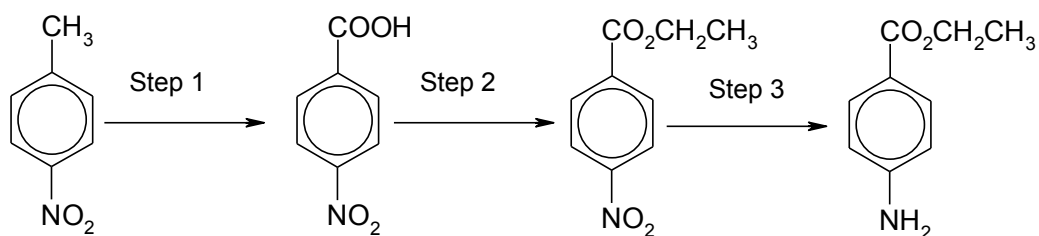
What of the following is likely to be the element?

- A** magnesium **B** sodium
C silicon **D** phosphorous

- 16** Which element is expected to show the greatest tendency to form some covalent compounds?

- A** barium
B calcium
C potassium
D beryllium

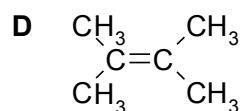
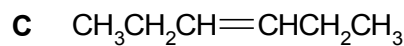
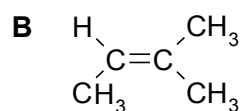
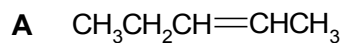
- 17** Many sunburn ointments contain benzocaine which relieves the pain caused by sunburn. It can be made in the laboratory by using the following reaction scheme.



Which of the following statements about this reaction scheme is **incorrect**?

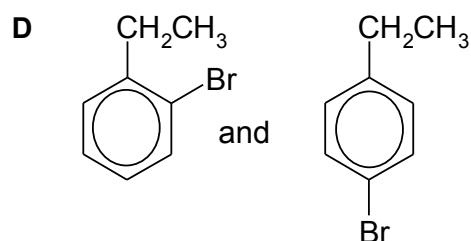
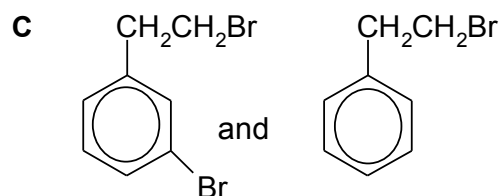
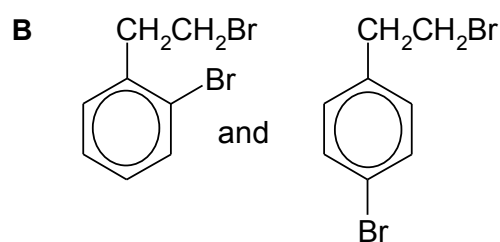
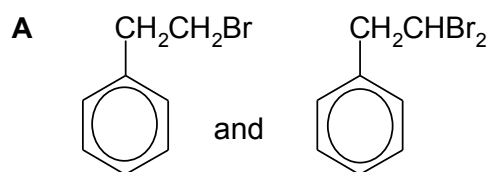
- A** Step 1 is an oxidation.
- B** Step 2 is an esterification.
- C** Step 3 is a substitution.
- D** Step 3 is a reduction.

- 18 Which hydrocarbon, on treatment with hot acidified potassium manganate(VII), would give propanoic acid only?



- 19 Iron (III) bromide is added to a solution containing equimolar quantities of ethylbenzene and bromine. The mixture is immediately placed in the dark and left until no further change takes place.

What are likely to be the main products?



- In which sequence are compounds listed in increasing order of their ODPs?

B $\text{CH}_3\text{CCl}_2\text{F}_2 < \text{CCl}_2\text{FCCl}_2\text{F}_2 < \text{CH}_3\text{CCl}_2\text{F}$

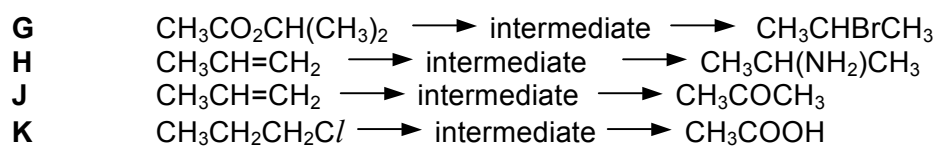
D $\text{CH}_3\text{CCl}_2\text{F} < \text{CCl}_2\text{FCClF}_2 < \text{CH}_3\text{CClF}_2$

- D** Both decolourises purple potassium manganate(VII).

- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- D** Butan-1-ol

24 Which pairs of reactions could have the same common intermediate?



A G and J

C G and K

B H and J

D H and K

25 Separation of ethylbenzene from a mixture of ethylbenzene and a carboxylic acid could involve

A Treating the mixture with dilute aqueous acid.

B Treating the mixture with dilute aqueous alkali.

C Extracting the carboxylic acid with cyclohexanol.

D Extracting the ethylbenzene with diethyl ether ($\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$).

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which statements about a 28.0 g sample of $^{14}\text{N}_2$ are correct?

- 1** The number of molecules is half the number of atoms in 32.0 g of $^{16}\text{O}_2$.
- 2** The number of atoms is twice the number of atoms in 4.0 g of ^4He .
- 3** The number of atoms is the same as the number of atoms in 24.0 g of ^{12}C .

27 Which of the following pairs do the molecules have similar shapes?

- 1** BCl_3 and AlCl_3
- 2** NO_2 and SO_2
- 3** BeF_2 and PF_3

28 Boron nitride is formed by the direct combination between boron and nitrogen when heated. It has a structure similar to graphite. It is likely to be

- 1** lubricant.
- 2** abrasive.
- 3** hard.

The responses **A** to **D** should be selected on the basis

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

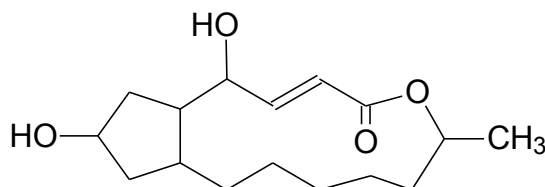
- 29** An attractant produced by butterflies and moths has the following structural formula:



Based on the above structure, which of the following is true of the attractant?

- 1** It can be hydrolysed by cold aqueous sodium hydroxide.
- 2** It decolourises hot acidified potassium manganate(VII) ions.
- 3** It could exist as a pair of geometric isomers.

- 30** Brefeldin A is an antibiotic.



Brefeldin A

Which of the following reagents react with Brefeldin A?

- 1** anhydrous PCl_5
- 2** hot acidified potassium dichromate(VII)
- 3** 2,4-dinitrophenylhydrazine

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